

Key Insight: Fusion-Space Reconstruction

Prior Work

Raw Feature Space

VAE/Diffusion reconstructs
768d raw text from A+V (196d)

- [x] Large networks (10-50x params)
- [x] Heterogeneous dimensions
- [x] Hard optimization

CVAE-MSA (Ours)

Fusion Space (40d)

Single CVAE reconstructs
40d fusion embedding from A+V (80d)

- [v] Lightweight (+30K params, +8.8%)
- [v] Unified representation space
- [v] KL weight explains 63% of variance

+30K

Extra
params

63%

MissT variance
explained by KL

0.597

Best MissT
(beats concat +1pp)

57

Systematic
experiments

KL Regularization is the Dominant Factor

